

Homework 1 Solutions

Vectors, Matrices, and Linear Systems

Instructor worked solution. Every step below is the level of work expected from students.

Problem 1 (5 pts)

$$u = [2, -1, 3]^T, \quad v = [1, 4, -2]^T, \quad w = [-4, 2, 6]^T$$

(a) $(2u - v) \cdot w$

$$2u - v = [4, -2, 6]^T - [1, 4, -2]^T = [3, -6, 8]^T$$

$$(2u - v) \cdot w = (3)(-4) + (-6)(2) + (8)(6) = -12 - 12 + 48 = 24$$

Final answer: 24

(b) $\|v + (1/2)w\|$

$$(1/2)w = [-2, 1, 3]^T$$

$$v + (1/2)w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T$$

$$\|v + (1/2)w\| = \sqrt{(-1)^2 + 5^2 + 1^2} = \sqrt{1 + 25 + 1} = \sqrt{27} = 3\sqrt{3} \approx 5.20$$

Final answer: $3\sqrt{3}$ (≈ 5.20)

(c) Angle between u and w

$$\text{The angle } \theta \text{ satisfies } \cos \theta = (u \cdot w) / (\|u\| \|w\|).$$

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$\|u\| = \sqrt{4 + 1 + 9} = \sqrt{14}$$

$$\|w\| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$\cos \theta = 8 / (\sqrt{14} \cdot 2\sqrt{14}) = 8 / 28 = 2/7$$

$$\theta = \arccos(2/7) \approx 73.40^\circ$$

Final answer: $\theta \approx 73.40^\circ$

Problem 2 (6 pts)

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & -2 \\ 0 & 4 & 1 \end{bmatrix} \quad (3 \times 3), \quad B = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & 1 \end{bmatrix} \quad (2 \times 3), \quad v = [1, -2, 3]^T \quad (3 \times 1), \quad w = [2, 0, -1]^T \quad (3 \times 1)$$

(a) AB is undefined: A has 3 columns but B has 2 rows.

BA is 2×3 :

$$\text{row 1: } [1(2)+0(1)+(-2)(0), \quad 1(-1)+0(3)+(-2)(4), \quad 1(0)+0(-2)+(-2)(1)] = [2, -9, -2]$$

$$\text{row 2: } [3(2)+1(1)+1(0), \quad 3(-1)+1(3)+1(4), \quad 3(0)+1(-2)+1(1)] = [7, 4, -1]$$

$$BA = \begin{bmatrix} 2 & -9 & -2 \\ 7 & 4 & -1 \end{bmatrix}$$

(b) $Av = [2(1)+(-1)(-2)+0(3), \quad 1(1)+3(-2)+(-2)(3), \quad 0(1)+4(-2)+1(3)]^T = [4, -11, -5]^T$

vA is undefined: v has 1 column but A has 3 rows.

(c) Aw is undefined: A has 3 columns but w has 1 row.

$$wA = [2(2)+0(1)+(-1)(0), \quad 2(-1)+0(3)+(-1)(4), \quad 2(0)+0(-2)+(-1)(1)] = [4 \quad -6 \quad -1]$$

(d) vw is the 3x3 outer product:

$$\begin{aligned} vw &= \begin{bmatrix} 1(2) & 1(0) & 1(-1) \\ -2(2) & -2(0) & -2(-1) \\ 3(2) & 3(0) & 3(-1) \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ -4 & 0 & 2 \\ 6 & 0 & -3 \end{bmatrix} \end{aligned}$$

wv = 2(1) + 0(-2) + (-1)(3) = -1 (a 1x1 matrix; this is the dot product of w and v)

(e) A² = AA

$$\text{row 1: } [2(2)+(-1)(1)+0(0), \quad 2(-1)+(-1)(3)+0(4), \quad 2(0)+(-1)(-2)+0(1)] = [3, -5, 2]$$

$$\text{row 2: } [1(2)+3(1)+(-2)(0), \quad 1(-1)+3(3)+(-2)(4), \quad 1(0)+3(-2)+(-2)(1)] = [5, 0, -8]$$

$$\text{row 3: } [0(2)+4(1)+1(0), \quad 0(-1)+4(3)+1(4), \quad 0(0)+4(-2)+1(1)] = [4, 16, -7]$$

$$A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}$$

(f) B² is undefined: B is 2x3, not square, so BB requires 3 = 2, which fails.

Problem 3 (6 pts)

(a) Augmented matrix:

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 2 & 3 & 1 & | & 2 \\ -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$\begin{aligned} R2 \rightarrow R2 - 2R1: & \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ -1 & 1 & 2 & | & -1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} R3 \rightarrow R3 + R1: & \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} R2 \rightarrow -R2: & \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} R3 \rightarrow R3 - 3R2: & \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 10 & | & 10 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} R3 \rightarrow (1/10)R3: & \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 1 & | & 1 \end{bmatrix} \end{aligned}$$

Back substitution: $x_3 = 1$; $x_2 = -4 + 3(1) = -1$; $x_1 = -1 - 2(-1) + 1 = 2$.
The system has exactly one solution.

Final answer: $x = [x_1, x_2, x_3]^T = [2, -1, 1]^T$

(b) Augmented matrix:

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 2 & 5 & 1 & | & 6 \\ 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_1: \\ \begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \\ 3 & 7 & 0 & | & 10 \end{bmatrix} \end{array}$$

$$\begin{array}{l} R_3 \rightarrow R_3 - 3R_1: \\ \begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \\ 0 & 1 & 3 & | & -2 \end{bmatrix} \end{array}$$

$$\begin{array}{l} R_3 \rightarrow R_3 - R_2: \\ \begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \end{array}$$

$$\begin{array}{l} R_1 \rightarrow R_1 - 2R_2: \\ \begin{bmatrix} 1 & 0 & -7 & | & 8 \\ 0 & 1 & 3 & | & -2 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \end{array}$$

The third column has no pivot, so z is a free variable. Let $z = t$.

$x = 8 + 7t$, $y = -2 - 3t$, $z = t$.

There are infinitely many solutions.

Final answer: $[x, y, z]^T = [8, -2, 0]^T + t [7, -3, 1]^T$, $t \in \mathbb{R}$

Problem 4 (8 pts)

(a) Let x_1, x_2, x_3 be the masses (in kg) of Alloy 1, Alloy 2, and Alloy 3 used. The product is 100 kg, so it contains 36 kg copper, 44 kg zinc, and 20 kg nickel.

Copper: $0.5 x_1 + 0.2 x_2 + 0.4 x_3 = 36$

Zinc: $0.3 x_1 + 0.7 x_2 + 0.2 x_3 = 44$

Nickel: $0.2 x_1 + 0.1 x_2 + 0.4 x_3 = 20$

(Adding the three equations gives $x_1 + x_2 + x_3 = 100$, as expected.)

(b) Multiplying each equation by 10 to clear decimals, the augmented matrix is

$$\begin{bmatrix} 5 & 2 & 4 & | & 360 \\ 3 & 7 & 2 & | & 440 \\ 2 & 1 & 4 & | & 200 \end{bmatrix}$$

$$\begin{array}{l} (c) R_1 \rightarrow R_1 - 2R_3: \\ \begin{bmatrix} 1 & 0 & -4 & | & -40 \\ 3 & 7 & 2 & | & 440 \\ 2 & 1 & 4 & | & 200 \end{bmatrix} \end{array}$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 3R_1: \\ \begin{bmatrix} 1 & 0 & -4 & | & -40 \\ 0 & 7 & 14 & | & 560 \\ 2 & 1 & 4 & | & 200 \end{bmatrix} \end{array}$$

$$\begin{array}{l} R_3 \rightarrow R_3 - 2R_1: \\ \begin{bmatrix} 1 & 0 & -4 & | & -40 \\ 0 & 7 & 14 & | & 560 \\ 0 & 1 & 12 & | & 280 \end{bmatrix} \end{array}$$

$$\begin{array}{l} R_2 \rightarrow (1/7)R_2: \\ \begin{bmatrix} 1 & 0 & -4 & | & -40 \\ 0 & 1 & 2 & | & 80 \\ 0 & 1 & 12 & | & 280 \end{bmatrix} \end{array}$$

$$\begin{array}{lcl}
 & & \begin{bmatrix} 0 & 1 & 12 & | & 280 \\ 1 & 0 & -4 & | & -40 \\ 0 & 1 & 2 & | & 80 \\ 0 & 0 & 10 & | & 200 \end{bmatrix} \\
 R3 \rightarrow R3 - R2: & & \\
 & & \begin{bmatrix} 0 & 1 & 12 & | & 280 \\ 1 & 0 & -4 & | & -40 \\ 0 & 1 & 2 & | & 80 \\ 0 & 0 & 10 & | & 200 \end{bmatrix} \\
 R3 \rightarrow (1/10)R3: & & \\
 & & \begin{bmatrix} 0 & 1 & 12 & | & 280 \\ 1 & 0 & -4 & | & -40 \\ 0 & 1 & 2 & | & 80 \\ 0 & 0 & 1 & | & 20 \end{bmatrix}
 \end{array}$$

Back substitution: $x_3 = 20$; $x_2 = 80 - 2(20) = 40$; $x_1 = -40 + 4(20) = 40$.

Check: $40 + 40 + 20 = 100$ kg. Copper: $20 + 8 + 8 = 36$ kg. ✓

Final answer: 40 kg of Alloy 1, 40 kg of Alloy 2, 20 kg of Alloy 3.

Problem 5 (7 pts)

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$$

$$\begin{array}{lcl}
 \text{(a) } R2 \rightarrow R2 - 2R1: & & \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 3 & 6 & 4 \end{bmatrix} \\
 & & \\
 R3 \rightarrow R3 - 3R1: & & \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \\
 & & \\
 R3 \rightarrow R3 - R2: & & \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \\
 & & \\
 R1 \rightarrow R1 - R2: & & \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}
 \end{array}$$

$$\text{Final answer: } \text{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

(b) $\text{rank}(A) = 2$, because the RREF has two pivot positions (columns 1 and 3).

(c) Zero solutions: possible. Row-reducing $[A \mid b]$ with the same operations gives a third row

of $[0 \ 0 \ 0 \mid b_3 - b_1 - b_2]$. If $b_3 \neq b_1 + b_2$ (for example $b = [0, 0, 1]^T$), the last row reads

$0 = \text{nonzero}$, so the system is inconsistent.

Exactly one solution: impossible. Column 2 is never a pivot column, so whenever the system is

consistent, x_2 is a free variable and there are infinitely many solutions.

Equivalently,

$\text{rank}(A) = 2 < 3 = \text{number of unknowns}$.

Infinitely many solutions: possible. If $b_3 = b_1 + b_2$ (for example $b = 0$, or $b = [1, 3, 4]^T$),

the system is consistent and has a free variable, so there are infinitely many solutions.

Problem 6 (8 pts)

(a) $v_1 = [1, 0, 2]^T$, $v_2 = [2, 1, 3]^T$, $v_3 = [0, 1, -1]^T$

Row reduce $[v_1 \ v_2 \ v_3]$:

$$\begin{array}{rclcl}
 & [1 & 2 & 0] & R_3 \rightarrow R_3 - 2R_1 & [1 & 2 & 0] & R_3 \rightarrow R_3 + R_2 & [1 & 2 & 0] \\
 R_1 \rightarrow R_1 - 2R_2 & [1 & 0 & -2] & & & & & & \\
 & [0 & 1 & 1] & \xrightarrow{\hspace{1cm}} & [0 & 1 & 1] & \xrightarrow{\hspace{1cm}} & [0 & 1 & 1] \\
 \xrightarrow{\hspace{1cm}} & [0 & 1 & 1] & & & & & & \\
 & [2 & 3 & -1] & & [0 & -1 & -1] & & [0 & 0 & 0] \\
 [0 & 0 & 0] & & & & & & &
 \end{array}$$

Only two pivots for three vectors, so the set is linearly dependent.

In fact $v_3 = v_2 - 2v_1$: $[2, 1, 3]^T - [2, 0, 4]^T = [0, 1, -1]^T$. ✓

v_1 and v_2 are not parallel, so the span is a plane through the origin in $\mathbb{R}^3$.

Final answer: linearly dependent; span is a plane.

(b) $v_1 = [1, 1, 0]^T$, $v_2 = [0, 1, 1]^T$, $v_3 = [1, 0, 1]^T$

Row reduce $[v_1 \ v_2 \ v_3]$:

$$\begin{array}{rclcl}
 [1 & 0 & 1] & R_2 \rightarrow R_2 - R_1 & [1 & 0 & 1] & R_3 \rightarrow R_3 - R_2 & [1 & 0 & 1] \\
 [1 & 1 & 0] & \xrightarrow{\hspace{1cm}} & [0 & 1 & -1] & \xrightarrow{\hspace{1cm}} & [0 & 1 & -1] \\
 [0 & 1 & 1] & & [0 & 1 & 1] & & [0 & 0 & 2]
 \end{array}$$

Three pivots for three vectors, so the only solution to $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ is

$c_1 = c_2 = c_3 = 0$. The set is linearly independent.

Three independent vectors in $\mathbb{R}^3$ span all of $\mathbb{R}^3$.

Final answer: linearly independent; span is all of $\mathbb{R}^3$.